

RITIK KUMAR

5th-year, BS-MS (Research) Physics | Indian Institute of Science, Bengaluru

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homepage

PROFILE

Physics researcher at the Indian Institute of Science working in machine learning, statistical analysis, scientific computing, and large-scale data analysis. I have worked on developing machine-learning and probabilistic models for noisy and irregular time-series data, including Deep Gaussian Processes, ReFANN, Siamese networks, and Physics-Informed Neural Networks. My work involves model development, optimisation, uncertainty estimation, validation, and working with large datasets. I also have experience in astrophysical data analysis using XMM-Newton and NuSTAR. Strong background in probability, statistics, linear algebra, optimisation, and numerical methods, with most of my work done in Python. Strong quantitative and mathematical foundations (JEE Advanced AIR 3424, top 0.3%).

EDUCATION

Indian Institute of Science (IISc), Bengaluru

Aug 2022 – July 2027 (*expected*)

BS-MS (Research) in Physics

CGPA: 7.5/10

- Linear Algebra, Multi-variable Calculus, Probability & Statistics, Machine Learning, Matrix Theory, Stochastic models & Application, Mathematical Methods in Physics, Computational Physics. Radiative Processes, Galaxies & ISM, General Relativity & Cosmology, Astronomical Techniques, Fluid Mechanics & Plasma Physics

TECHNICAL SKILLS

- **ML / DL:** PyTorch, GPyTorch, Deep Gaussian Processes, ReFANN, Siamese Networks, PINNs, Bayesian Neural Networks, MC Dropout, uncertainty estimation
- **Quantitative / Statistical:** Probability and Statistics, Statistical Inference, Time-Series Analysis, Bayesian Inference, Quantile Regression, Cross-Validation, Bootstrap Resampling, Robust Statistics, ablation studies, model comparison (χ^2 , AIC/BIC).
- **Math:** probability, linear algebra, optimisation, numerical methods, stochastic models
- **Programming:** Python (NumPy, SciPy, Pandas, Astropy, scikit-learn, Matplotlib), Bash, some C, LaTeX.
- **Computing** HPC/SLURM, SSH, Parallel Processing, XMM-SAS, NuSTARDAS, XSPEC/ISIS

PUBLICATIONS

Publications

- Ayush, **Ritik**, Dainotti et al. (2026). Physics-Informed Neural Networks and Data-Driven Models for GRB X-ray Light-Curve Gap Reconstruction. [2026](#). *arXiv pre-print, paper is in peer-review*.
- Kaushal, A., Dainotti, M.G., **Ritik**, et al. (2026). Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves. *Journal of High Energy Astrophysics (JHEAp)*. [2026](#)
- **Ritik**, Stalin, C.S., Reeves, J.N. Possible detection of unambiguous outflows in a Luminous Quasar. Submitted to ApJ Journal, 2026.

In Preparation

Ritik, Stalin, C.S., Hota, J. Variation in Coronal Temperature in Seyfert-type AGN. In preparation, 2026.

RESEARCH & ML EXPERIENCE

ML Researcher, Time-Series Reconstruction

Apr 2025 – July 2026

National Astronomical Observatory of Japan (NAOJ) Advisor: Prof. Maria G. Dainotti

- **2nd author** (co-corresponding), Ayush, **Ritik**, Dainotti et al. ([2026-arxiv pre-print](#)) and it is in (peer-review in journal): • Benchmarked PINN, ReFANN, Siamese networks, and polynomial quantile regression on 545 noisy, irregularly sampled time series under a common evaluation pipeline, comparing predictive accuracy, uncertainty, and robustness.

- Implemented uncertainty-aware neural networks in PyTorch (mean and log-variance outputs, Gaussian NLL loss, MC dropout). **Siamese network (own reference-anchor fusion design):** -22.9% / -22.8% / -26.9% **fitted-parameter uncertainty, with $\leq 3.7\%$ outliers. ReFANN:** -19.8% / -21.4% / -26.5% **on 522 series.**
- Co-authored on quantifying a **gain-versus-robustness trade-off**: the PINN's best prior cut uncertainty by $\sim 41\text{--}49\%$ but produced up to $\sim 17\%$ outliers, versus 4% for the data-driven models.
- Implemented uncertainty-aware neural networks in PyTorch using mean and log-variance outputs, Gaussian NLL loss, and MC dropout; evaluated predictive uncertainty and model robustness alongside point estimates.
- Fixed a PINN training collapse (derivative-based physics loss converged to a constant output) using a value-space loss, λ warm-up, EMA loss normalisation and automatic restart.
- Tuned the PINN with Optuna (80 trials per series, 5-fold CV objective); validated Siamese and ReFANN with 5-fold CV and a λ -sensitivity ablation.
- **Co-author, 1. Kaushal, Dainotti, Ritik et al. (JHEAp 51, 100519 (2026)):** two-layer variational Deep GP (GPyTorch), one of seven models compared on 521 series.

ML Researcher – Physics-Informed Neural Networks (PINNs)

Jan 2025 – Apr 2025

IISc Bengaluru Advisor: Prof. Rishita Das (Aerospace Engineering)

- Implemented PINNs in PyTorch to solve 1D, 2D, and 3D Burgers' equations and 1D Euler equations, including the Sod shock-tube problem with sharp discontinuities.
- Compared the results with Runge-Kutta, Euler, central-difference methods, used CFL criterion for stable time-stepping.

Lead Researcher – AGN Spectral Modelling (ECAP Collaboration)

Jan 2026 – Present

Erlangen Centre for Astroparticle Physics (ECAP), FAU Germany Full Group Member

- Working with XMM-Newton and NuSTAR multi-epoch data for X-ray spectral analysis of active galactic nuclei.
- Perform statistical model fitting, MCMC inference, and model comparison on multi-epoch XMM-Newton and NuSTAR data; develop Python analysis pipelines and run computationally intensive workloads on HPC using parallel processing.

Tracing of Outflows in Active Galactic Nuclei – Bachelor's Thesis

Sep 2025 – Apr 2026

Indian Institute of Astrophysics (IIA) Advisor: Prof. C. S. Stalin

- Built a Python pipeline to cross-match about 1 million AGN catalogue entries with XMM-Newton observations and automate filtering and spectral extraction.
- Used 23-worker parallel processing and developed custom XSTAR grids to study complex X-ray absorption features and multi-phase ultra-fast outflows. **Submitted to Journal and currently is in peer-review.**

Spectral Analysis of Seyfert Galaxies

Dec 2024 – Jan 2026

Indian Institute of Astrophysics (IIA) Advisor: Prof. C. S. Stalin

- Developed a Python pipeline to cross-match 30,000 sources, perform data reduction, background and S/N selection.
- Worked with NuSTAR and XMM multi-epoch observations and performed broadband spectral modelling in XSPEC.
- Used X-ray spectral modelling to study and measure coronal temperatures in galaxies. **Manuscript in preparation.**

CONFERENCES & WORKSHOPS

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- XRISM International Conference 2025, Japan – [Certificate](#).
 - Astronomical Society of India Conference 2026, Guwahati – ([Certificate](#), [Poster](#)).

ACHIEVEMENTS

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- **Internship, NAOJ Tokyo (Summer 2025)** and **collaboration** with Prof. Maria G. Dainotti since April 2024.
 - **Full membership, ECAP X-ray Research Group, Germany** since Jan. 2026
 - **Research collaboration with the Indian Institute of Astrophysics (IIA), Bengaluru** since Dec. 2024.
 - **Management Coordinator, Pravega 2024–25** (India's largest science & cultural festival), IISc